

Algorithmic Feed Simulation Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Technology • 30-40 min

Category	Knowledge and Technology	Time	30-40 min
Difficulty	Medium	ID	algorithmic-feed-simulation

Big idea: A feed is a knowledge filter.

Overview

Students experience how personalisation filters knowledge.

Student Prompt

Inspect Algorithmic Feed Simulation Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Algorithmic Feed Simulation Stimulus A	Student-facing stimulus 1: Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Algorithmic Feed Simulation Stimulus B	Student-facing stimulus 2: A feed that changes after each student 'likes' or skips a card. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Algorithmic Feed Simulation Reveal / Twist	Student-facing stimulus 3: A comparison between a curiosity-maximising feed and a reliability-maximising feed. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — A feed is a knowledge filter.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Algorithmic Feed Simulation Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Algorithmic Feed Simulation Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Do algorithms expand knowledge or narrow it?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how a feed is a knowledge filter.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Paper post cards ranked by engagement, recency, outrage, or similarity to past clicks.
- A feed that changes after each student 'likes' or skips a card.
- A comparison between a curiosity-maximising feed and a reliability-maximising feed.

Suggested Timing

Stage	Teacher move	Watch for
1	Students secretly choose five cards they would click.	Assumptions, confidence shifts, evidence claims, and language choices.
2	The teacher acts as the algorithm and gives each student more cards similar to their choices.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Repeat for three rounds.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Students compare how different feeds became over time.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may treat a tool's output as neutral because it feels automatic.
- Students may focus only on user error and miss design incentives or hidden rules.

Assessment Look-Fors

- Identifies the rule, design choice, or incentive shaping knowledge.
- Distinguishes access to information from justified knowledge.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Search Engine Ranking Game
- AI Hallucination Courtroom
- Deepfake Detection Challenge